ISTANBUL TECHNICAL UNIVERSITY

COMPUTER ENGINEERING DEPARTMENT

BLG 435E
Artificial Intelligence

Homework 2

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Contents

1 Metric results and observations **1**

2 Why certain models performed better **1**

3 Effect of scaling and hyperparameter tuning **1**

4 One limitation and one improvement idea **1**

REFERENCES **3**

1 Metric results and observations

For Regression,

- Linear regression achieve reasonable performance with scaling.
- Ridge regression shows a bit better performance through regularization.

For Classification,

- Logistic regression achieve 98% accuracy with 99% F1 score, which is really impressive.
- Random forest methods result was also impressively nice, but not better than Logistic regression.

Because of the size of the dataset, the complex models' improve on the prediction was not observed clearly.

2 Why certain models performed better

In the regression task, Ridge regression performed better thanks to the regularization effect that reduced overfitting on the relatively small and feature-limited diabetes dataset.

The classification models performed almost the same because the breast cancer dataset was clean and naturally separable and had strong features that allowed for easy differentiation between malignant and benign cases.

3 Effect of scaling and hyperparameter tuning

Scaling was important because logistic regression is sensitive to feature size, and standardizing the inputs improved its stability and convergence. Linear regression also benefits from features being at comparable scales, producing more reliable coefficients, while Random Forest, thanks to its tree-based structure that does not rely on feature distances, was largely unaffected.

Hyperparameter tuning in both the regression models and the tree-based classifier allowed the algorithms to automatically select the most appropriate parameter settings. By searching through candidate values, the models could adjust their levels of complexity and regularization to achieve the best possible performance. This process ensured that each model was configured in the most efficient way, rather than relying on default parameters.

4 One limitation and one improvement idea

The models achieved suspiciously similar performance on classification, likely because the dataset was both very clean and relatively small, making it easy for most algorithms to handle. This reduces the variability in model behavior and limits our ability to meaningfully compare their strengths and weaknesses.

To obtain more informative results, the dataset could be expanded or augmented to introduce more variation and difficulty. Additionally, using more diverse algorithms such as SVM or Gradient Boosting, as well as ensemble methods like a voting classifier, would help increase model diversity.

REFERENCES